Causal estimation of the ATE on observational data: comparison between AIPW and TMLE

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Short description. This project compares two doubly robust estimators of the Average Treatment Effect, AIPW and TMLE (with machine-learning and deep-learning outcome models), on the LaLonde / National Supported Work data combined with PSID/CPS controls. It evaluates accuracy against the experimental benchmark, efficiency (standard error) and behavior under poor overlap.

1. Introduction

1.1 Causal inference

Causal inference deals with estimating the effect of a treatment T on an outcome Y, distinguishing the observed association from the underlying causal relationship. In the potential outcomes framework (Rubin, 1974), each unit i has two potential outcomes, $Y_i(1)$ and $Y_i(0)$, of which only one is observed (the fundamental problem of causal inference). The average treatment effect is defined as

$$tau_{ATE} = E[\ Y(1) - Y(0)\ ].$$

Under the standard assumptions of unconfoundedness (Y(0), Y(1) independent of T given X) and positivity (0 < e(X) < 1), the ATE is identifiable from observational data.

1.2 Application context

With observational data, treatment assignment is not randomized: the raw comparison between group means is biased by selection bias. Adjustment techniques (matching, weighting, outcome modeling) aim to remove this bias by reconstructing a valid counterfactual.

1.3 Project objective

The objective is to compare two pipelines for the post-hoc estimation of the ATE — Augmented Inverse Probability Weighting (AIPW) and Targeted Maximum Likelihood Estimation (TMLE) — in order to identify the most robust one for the dataset at hand, in terms of accuracy (deviation from the experimental benchmark), efficiency (standard error) and behavior under poor overlap.

1.4 Dataset

The estimation was carried out on the National Supported Work (NSW, LaLonde 1986) sample, which by construction exhibits strong overlap between the treated and control distributions, since it comes from a randomized experiment. The analysis was then extended to an observational dataset, obtained by combining the treated population of the NSW sample (185 units) with the Current Population Survey/PSID control group (about 15,000 units), in order to reproduce realistic conditions of selection bias and poor overlap between the treated and untreated populations.

1.5 Libraries

- **numpy, pandas:** data handling and manipulation, the de facto standard.
- **cem:** Coarsened Exact Matching, for pre-balancing the combined dataset and building matched subsamples.
- **scikit-learn:** definition, fitting, preprocessing, model selection and metrics for the outcome/propensity models.
- **keras:** implementation of the neural networks used as outcome model in the deep variant of TMLE.
- **PyALE:** computation of Accumulated Local Effects to interpret interactions between covariates in the outcome models.

2. Core of the analysis

2.1 Doubly Robust framework

Given the difficulty of the combined dataset (marked selection bias, high variance), a doubly robust (DR) framework was adopted, which guarantees consistency of the ATE estimate if at least one of the propensity model e-hat(X) and the outcome model mu-hat(X) is correctly specified. Within this framework, two estimators were compared, AIPW and TMLE, which, while sharing the double robustness property, differ in how they combine the two components.

2.1.1 AIPW

The AIPW estimator is defined as

$$\tau\text{-hat}_{AIPW} = (1/N)\,\Sigma_i\,[\,\mu\text{-hat}_1(X_i) + T_i\,(Y_i - \mu\text{-hat}_1(X_i))\,/\,e\text{-hat}(X_i)\,] - (1/N)\,\Sigma_i\,[\,\mu\text{-hat}_0(X_i) + (1 - T_i)\,(Y_i - \mu\text{-hat}_0(X_i))\,/\,(1 - e\text{-hat}(X_i))\,],$$

where $\mu\text{-hat}_1$ and $\mu\text{-hat}_0$ are the predictions of an outcome model fitted on the treated and control subsets respectively, and e-hat(X) = P-hat(T = 1 | X) is the propensity score. The correction term based on e-hat(X) gives more weight to control units whose covariates are similar to those of the treated, mitigating the residual selection bias of the outcome model alone.

2.1.2 TMLE

The Targeted Maximum Likelihood Estimator shares double robustness with AIPW — consistency if mu-hat(X, T) or e-hat(X) is correctly specified — but differs in its architecture: instead of applying an additive IPW correction a posteriori, TMLE incorporates the propensity score directly into the outcome predictions through a targeted clever covariate

$$H(T, X) = T\,/\,e\text{-hat}(X) - (1 - T)\,/\,(1 - e\text{-hat}(X)).$$

The initial predictions $\mu\text{-hat}^{(0)}$ are updated through a fluctuation step (targeting step), yielding a proper substitution estimator. Although originally formulated for binary outcomes, TMLE extends naturally to continuous variables by linearly rescaling the target to [0, 1] and using a logit link, respecting the natural bounds of the outcome and offering greater finite-sample efficiency in machine-learning-based pipelines.

2.1.3 Propensity Score

The propensity score e(x) = P(T = 1 | X = x) is the conditional probability of treatment assignment, fundamental to correcting selection bias. For its estimation, a generative approach was adopted, modeling the covariate densities P(X | T = t) with Gaussian Mixture Models (GMM). Using the learned distributions, the propensity score was finally computed as the posterior probability via Bayes' theorem:

$$e(x) = P(X|T=1)\,P(T=1) \,/\, [\, P(X|T=1)\,P(T=1) + P(X|T=0)\,P(T=0)\,] \quad (2.1)$$

2.2 Dataset and matching

The analysis was carried out both on the experimental NSW sample and on the combined NSW-CPS/PSID dataset, to which a preliminary matching was applied.

2.2.1 Coarsened Exact Matching

The ATE is, by definition, an estimand that requires comparison with each unit's counterfactual, which is not directly observable; the nearest unit in terms of covariates is therefore used as a proxy for the counterfactual, hence the need for matching. Coarsened Exact Matching (CEM) discretizes (coarsening) the covariates into bins and performs exact matching within each bin, guaranteeing exact balance (L1 distance) on the coarsened variables. The binning was defined following the reference literature on the NSW dataset.

2.3 Training

For the outcome model, Random Forest and Gradient Boosting were mainly used, chosen because they are widely used in the causal inference literature and performed better, on the dataset at hand, than alternatives tested in preliminary empirical analyses; no preprocessing was applied to the covariates for these tree-based models.

AIPW. The propensity score is estimated greedily, without parametric assumptions on the data distribution, optimizing overlap and BIC; the clever covariate H is then computed. The treated/control subset determines the choice of the most suitable outcome model via nested cross-validation.

TMLE (ML). The data are preprocessed by scaling Y with Min-Max scaling; nested cross-validation is carried out on the whole dataset. Three outcome predictions are estimated — Y-hat, Y-hat$_1$, Y-hat$_0$ — on the full sample, on the T = 1 subset and on the T = 0 subset respectively; finally, the offset for the adjustment (fluctuation) of the final estimates is computed.

TMLE (Deep Learning). Compared with the ML variant, the main change is the outcome model: Y is scaled with Min-Max, X with standardization (StandardScaler). Train/test/validation sets are built, with the training set enlarged through bootstrap sampling (2 replicates). A fully-connected network with dropout is used to limit overfitting, trained by minimizing the MSE and validated on the MAE; the rest of the pipeline (clever covariate, fluctuation) is analogous to TMLE-ML.

2.4 Results

Table 2.1: Comparison of ATE/ATT estimates on the combined NSW-PSID dataset

Method	Estimate	SE	95% CI
AIPW (ATE)	384.63	367.88	[-336.41, 1105.66]

TMLE (ML)	194.30	356	[-503, 892]
TMLE (Deep Learning)	1773.69	190	[1401, 2147]
Experimental benchmark	1794.34	-	-

The TMLE variant with a neural-network outcome model (Deep Learning) comes closest to the experimental benchmark (1794.34) and has the smallest standard error, making it the most efficient pipeline among those tested. AIPW, while remaining within the doubly robust framework, shows high variance (SE = 378.66) and an ATE estimate far from the benchmark, presumably because the tree-based outcome model cannot automatically capture the nonlinear interactions between covariates; the neural network, through its activation functions, builds more informative intermediate representations able to model such interactions. It should also be noted that, in none of the tested configurations does the confidence interval exclude a null or negative effect, a sign of the persistent uncertainty due to poor overlap between treated and controls in the combined dataset.

3. Conclusions

The project compared two doubly robust estimators — AIPW and TMLE — for estimating the ATE on an observational dataset characterized by strong selection bias, obtained by combining the experimental NSW sample with the PSID/CPS control group. The TMLE pipeline with a deep learning outcome model proved to be the most accurate and efficient, coming close to the experimental benchmark with the lowest standard error; AIPW, although theoretically equivalent in terms of robustness, showed greater instability linked to the outcome model's lower ability to capture complex interactions between covariates. As future work, we propose moving beyond sequential approaches through neural architectures for the simultaneous estimation of outcome and propensity score, as well as adopting Graph Transformers to model the whole causal graph directly and natively.